



BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: *[Autonomous Flying Delivery Robot]*
by

[Group_No: 07]

[Section : 06]

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Abstract

The main aim of this project was to build a secure autonomous aerial delivery robot. We used Arduino Mega 2560 as the microcontroller and MultiWii 2.4 as the flight control firmware. Unfortunately, autonomous delivery could not be achieved. We were able to demonstrate flight at around 18m. The lock mechanism works perfectly with a separate ESP32 and a servo. This project can lead to a net positive impact in emergency delivery situations (flooded areas, remote areas) where terrestrial travel is impossible. The project has sustainability concerns due to the use of a LiPo battery (which could lead to waste). Other than that, the carbon footprint of the project is minimal.

Keywords

Arduino Mega 2560, MultiWii, Aerial Delivery Robot

1. Introduction

In emergency situations where terrestrial delivery is not possible, aerial delivery becomes one of the primary means of delivery. To that end, we wanted to build a secure autonomous aerial delivery robot. Our objectives were to

- Make a an autonomous delivery robot
 - Integrate Motors, ESCs, Propellers, and frame, and battery (and PDB) for a system that flies
 - Integrate GPS, compass, and bluetooth for waypoint navigation
 - Integrate ultrasonic sensor for obstacle detection and maintaining altitude
 - Integrate accelerometer/gyroscope for maintaining balance
- Make a lock mechanism
 - Integrate ESP32 and servo to make a lock mechanism using a box

The arduino mega 2560 interfaces with the rest of the system (computer interfacing).

2. Related Work/ Inspiration

Our project is inspired by medical delivery drones used in emergency healthcare services. Unlike large-scale commercial systems, this project emphasizes affordability, simplicity, and emergency usability rather than commercial efficiency. The focus on disaster-prone regions like Bangladesh differentiates our work, as it considers local social, safety, and cultural constraints often ignored in existing systems.

3. Technical Approach

System Architecture:

The main controller is arduino mega 2560. On top of that the multiwii architecture was used. And then all the relevant sensors were integrated.

Components used and cost breakdown



Webpage: www.robottechshop.com
Facebook Page: www.facebook.com/robottechshop
Address: Axion Engineering Solutions, House 36, Road 6, Block B, Banasree, Dhaka.

Shipping Address
N/A

Billing Address
Mahir Rafsan
BRACU

Invoice

Invoice Number 0220612016
Order Number 82177
Order Date 20/12/2025
Payment Method Cash on Delivery
Email rafsan5590@gmail.com
Telephone 01933792269

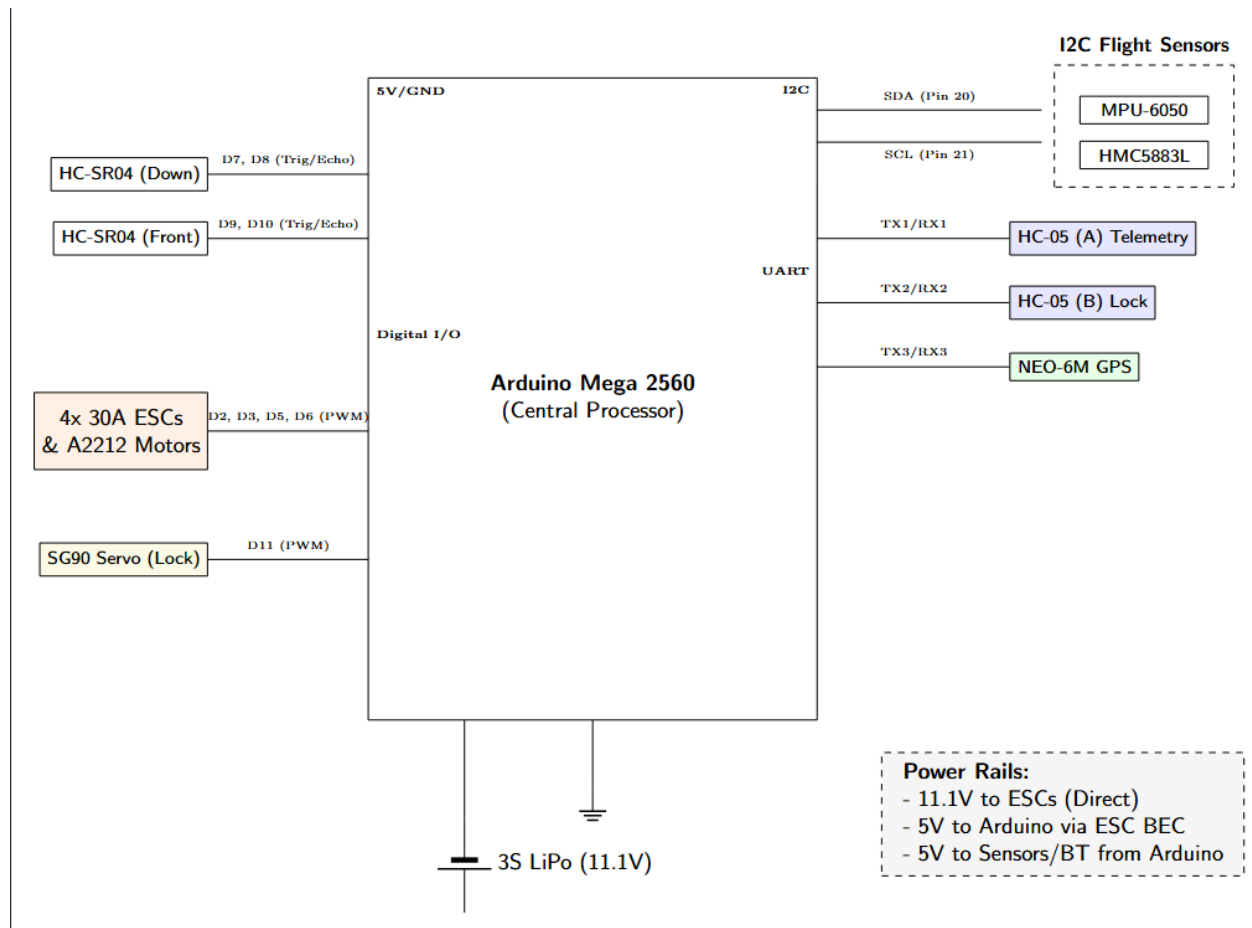
Product	Price	Quantity	Total
Arduino Mega 2560 ch340 driver with USB Cable SKU: s2-2-c1-62682	৳ 1,690	1	৳ 1,690
MPU 6500 6DOF Accelerometer and Gyroscope Module (6050 Upgraded) SKU: R2-4-C2-73094	৳ 165	1	৳ 165
Ultrasonic Ranging Module Sonar Sensor (HC-SR04) SKU: s1-4-c5-00045-400W	৳ 95	2	৳ 190
Brushless Motor A2212/13T 1000KV -for RC Drones & Airplanes SKU: D1-8-52365	৳ 520	4 - 2	৳ 2,080
30A ESC Brushless Motor Speed Controller SKU: s2-4-c8-65451	৳ 500	4	৳ 2,000
1550 mAh 3S 35C LiPower 11.1 V High-Discharge LiPo Battery with XT-60 Connector SKU: CA-14-34891	৳ 1,680	1	৳ 1,680
HC-05 Bluetooth Module For Arduino SKU: CA-7-00653-blm	৳ 255	2	৳ 510
GPS MODULE FOR ARDUINO NEO 6M UBLOX SKU: s2-9-sd4-01457	৳ 530	1	৳ 530
Silicone Wire 16AWG 16 Gauge Flexible Tinned Copper Wire -1ft Colors : BLUE SKU: 89435-04	৳ 30	3	৳ 90
Silicone Wire 16AWG 16 Gauge Flexible Tinned Copper Wire -1ft Colors : GREEN SKU: 89435-05	৳ 30	2	৳ 60
Male to Male Jumper wire Connector(20pc) SKU: r1-3-c3-00177	৳ 50	1	৳ 50
Male to Female Jumper wire Connector(20Pc) SKU: r1-3-c3-00181	৳ 50	1	৳ 50
10 inch Propeller Prop Blades Rotor for RC Multicopter Quadcopter Helicopter UAV SKU: CTN-01-76294	৳ 150	3	৳ 450
180 degree Micro Servo Motor 9g sg90 SKU: S2-10-servo-00364	৳ 112	1	৳ 112
Subtotal			৳ 9,880
Discount			
Shipping	29		৳ 9,909
Total			৳ 9,909

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Circuit:



Functionality:

1. Robot flies using radio telemetry and remote control
 - a. Mission plan is activated using remote control rx
2. The lock mechanism secures the payload using servo and ESP32

Challenges

- GPS module does not work. So, we could not implement autonomous navigation
- We could not integrate ultrasonic sensor, bluetooth, compass due to lack of time

4. Sustainability & Impact

Sustainability:

The robot has a minimum carbon footprint. The only concern is battery waste

Impact:

Socially, the robot can significantly improve access to emergency supplies in remote areas. In emergency situations, this robot can have a big impact where terrestrial delivery is not possible.

Future Work:

Future work includes finishing the current project plan. Further improvements include using a pixhawk for our flight controller. If more compute can be purchased, Vision-Language-Action language models can be used for genuine autonomous behaviour

Limitations:

- Payload weight limitations
- Dependence on weather conditions
- Regulatory restrictions on autonomous flight

5. Results & Discussion

The system successfully demonstrates the conceptual workflow of autonomous aerial delivery. This is a tough student project. We had to spend over 11,000 taka for this. Even then our project was not that good. Moreover, this project is very time consuming. It's not possible to do a good student project in this time frame. Moreover, there are minimal things to learn from this project. The only thing to do is to buy the parts and assemble them.

6. Conclusion

We just demonstrated using radio telemetry and rc control. We could not make the robot autonomous due to time constraints. In future, we will try to finish the project on our current project plan.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Mahir Muntasir Rafsan	22201104	Hardware Research	

			Component acquisition
Masudur Sadik Rifat	22201145	Sustainability Study	Poster composition
MD. Rakibul Hasan	22201569	Assembly	Building the entire system from scratch
Abu Safwan Md. Farhan	22201340	Project report	Project report writing
Fahim Bin Arif	22301537	Circuit drawing	Circuit diagram preparation
Suraaia Alam	22201175	Coordination	Coordination